

CENTRAL FALSIFIABLE SCIENTIFIC CLAIM:

"A fixed-size evolving state can carry useful information across sequences without storing every previous token, but increasing sequence length and conflicting updates can cause interference and information loss; additional inference-time computation can sometimes improve recovery."

1. Theoretical Foundations & Mechanics
- KV-Cache vs Bounded State:** Autoregressive Transformers maintain $O(T \cdot d)$ KV buffers. Bounded memory maintains $O(1)$ state M_t via outer product binding: $M_t = \lambda M_{t-1} + \Delta_t (v_t \otimes k_t^T)$.
 - Superposition & Capacity Limits:** Under orthogonal keys, retrieval is exact: $v^* = M_T^{-1} q$. When $N > \alpha_c \cdot d$ (Hopfield/Associative limit $\alpha_c \approx 0.14$), cross-talk noise causes graceful superposition interference.
 - Inference-Time Scaling (Latent Deliberation):** Test-time refinement optimizes query sharpening without updating model weights: $v^{(k+1)} = (1 - \eta) v^{(k)} + \eta M_T^{-1} q_{\text{sharp}}^{(k)}$, helping de-noise superimposed coordinates.
3. Empirical Sweep Results (2,700 Trials)
- Delayed Recall (Lag $k \in [4, 128]$):** Educational Toy maintains **100% accuracy** across all lags via selective Δ_t gating, whereas standard vector RNN decays to 0%.
 - Sequence Clutter ($T \in [32, 512]$):** Full History maintains 100% at $O(T)$ latency penalty (7.4ms $\rightarrow$ 61.1ms). Toy maintains $O(1)$ latency with mild noise accumulation.
 - Destructive Overwrites ($N \in [1, 8]$):** Recency dominance observed; overwrite saturation drops accuracy from 100% ($N \leq 2$) to $\sim 45\%$ ($N \geq 5$).
 - Subspace Capacity Stress (N_{pairs} vs $d=32$):** Abrupt phase transition at $N > 4$ pairs, matching theoretical associative rank limits.
 - Test-Time Recovery:** Additional inference effort ($K=1$ to 16) yields measurable recovery when signal is in superposition, but cannot recover fully overwritten state.

2. Architectural Taxonomy & Comparison

Architecture	State Space	Step Latency	Interference Behavior
Transformer (KV)	$O(T \cdot d)$ Unbounded	$O(T)$ per step	Zero (Exact storage)
Selective SSM (Mamba)	$O(d)$ Bounded	$O(1)$ Constant	Decay over long horizons
BDH (Dragon Hatchling)	$O(1)$ Bounded State	$O(1)$ Constant	Evolving associative state
BDH-CQ (Latent Reason)	$O(1)$ Bounded State	$O(K)$ Variable	Test-time latent deliberation

4. Evidence Classification Standard

- [PUBLISHED]** BDH $O(N)$ linear time & state dynamics (Kosowski et al., 2025).
- [PUBLISHED]** BDH-CQ in-context latent reasoning (Engdahl et al., 2026).
- [EMPIRICAL]** 2,700 seeded trials in data/precomputed/sweeps_master.csv.
- [EDUCATIONAL TOY]** $d=32$ fast-weight surrogate for browser experimentation.

CRITICAL BOUNDARIES, LIMITATIONS & SCIENTIFIC DISTINCTIONS:

1. Educational Toy vs Published BDH: Our browser toy is a pedagogical $d=32$ associative matrix surrogate. It does NOT implement official multi-billion parameter BDH weights. | **2. Biological Analogy:** Synapse/neuron mathematical duality is illustrative linear algebra; no biological claims are made. | **3. Live vs Precomputed:** Live experiments execute in PyTorch/TypeScript; heavy sweeps are labelled Precomputed with full seed transparency. | **4. Testable Hypothesis:** Tests whether test-time compute improves recovery without asserting universal scaling.

Primary References: [1] Kosowski et al. (2025) *The Dragon Hatchling*, arXiv:2509.26507. [2] Engdahl et al. (2026) *BDH-CQ: In-Context Learning with Recurrent Latent Reasoning*, arXiv:2608.09888. [3] Gu & Dao (2023) *Mamba*. [4] Snell et al. (2024) *Scaling LLM Test-Time Compute*. — **Code & Data:** Open Source MIT / Deterministic Seeded (PyTorch 2.0+)